

A Partial Extension of Ray's $B_{p,q}$ Non-Embedding Criterion

FA/Banach discovery run fa_banach_001

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1 Source and Target

The source paper is Samya Kumar Ray, “On Isometric Embedding $\ell_p^m \rightarrow S_\infty^n$ and Unique operator space structure”, arXiv:1911.00241. For $1 \leq p, q < \infty$, Ray writes

$$B_{p,q} = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^p + |z_2|^q < 1\}$$

and denotes by $\|\cdot\|_{B_{p,q}}$ the norm whose open unit ball is $B_{p,q}$. Remark 4.2 proves non-embedding into S_∞ for $2 < p, q < \infty$, records the positive triangular embedding for $B_{1,2}$, and states the conjecture that $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ embeds isometrically into S_∞ if and only if

$$(p, q) = (2, 2), \quad (p, q) = (1, 2), \quad \text{or} \quad (p, q) = (2, 1).$$

Note that we always have for $z_2 \neq 0$, $\|(z_1, z_2)\|_p > |z_1|$. Therefore, from equation (4.4), we obtain

$$(4.5) \quad \|z_1 T_3 + z_2 S_3\|_{\ell_2 \rightarrow \ell_2} = (|z_1|^p + |z_2|^q)^{\frac{1}{p}}, \text{ for all, } z_2 \neq 0.$$

But by continuity of norm, we have from equation (4.5)

$$\|z_1 T_3 + z_2 S_3\|_{\ell_2 \rightarrow \ell_2} = (|z_1|^p + |z_2|^q)^{\frac{1}{p}}, \text{ for all, } z_1, z_2 \in \mathbb{C}.$$

Therefore, we obtain a new isometric embedding $(z_1, z_2) \mapsto z_1 T_3 + z_2 S_3$ from ℓ_p^2 to $B(\ell_2)$ where T_3 is again a positive semi-definite diagonal matrix but now there are $n - 1$ number of 1's in the diagonal. Now, we can argue as in the proof of Theorem 1.1 to obtain a contradiction. This completes the proof of the theorem. $\square$

Remark 4.2. The proof of Theorem 1.2 indicates that Theorem 1.2 holds true for any two dimensional complex Banach space $(\mathbb{C}^2, \|\cdot\|)$ with the following properties.

- (1) For all $(z_1, z_2) \in \mathbb{C}^2$, $\|(z_1, z_2)\| = \|(\overline{z_1}, \overline{z_2})\|$.
- (2) For all $0 < c \leq 1$, the operator $(z_1, z_2) \mapsto (z_1, cz_2)$ is not a contraction from $(\mathbb{C}^2, \|\cdot\|)$ to ℓ_2^2 .
- (3) $\{(z_1, z_2) \in \mathbb{C}^2 : \|(z_1, z_2)\| > |z_1|\}$ is dense in $\mathbb{C}^2$.
- (4) $\|(1, 0)\| = \|(0, 1)\| = 1$.
- (5) $\|(1, z)\| \geq 1$ and $\|(w, 1)\| \geq 1$ for all $z, w \in \mathbb{C}$.

Note that for all $2 < p, q < \infty$, the Banach spaces $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ satisfies above properties since for any $2 < p, q < \infty$, we can find $2 < p_1, p_2 < \infty$ such that $\|\cdot\|_{\ell_{p_1}} \leq \|\cdot\|_{B_{p,q}} \leq \|\cdot\|_{\ell_{p_2}}$. Hence, there is no isometric embedding of $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ into the set of all compact operators for $2 < p, q < \infty$. However, $(\mathbb{C}^2, \|\cdot\|_{B_{1,2}})$ embeds isometrically into S_∞^2 via the embedding

$$(z_1, z_2) \mapsto z_1 I_2 + z_2 e_{12}.$$

We conjecture that $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ embeds isometrically into S_∞ if and only if $(p, q) = (2, 2), (1, 2)$ or $(2, 1)$.

Proof of Theorem 1.3. If S_Γ^m embeds isometrically into S_p , for some $m \geq 2$, we can observe ℓ_1^2 embeds into S_p isometrically. Let $(z_1, z_2) \mapsto z_1 T + z_2 S$ be an isometry from ℓ_1^2 into S_p . Clearly, we have $T \parallel S$. Therefore, by Theorem 2.3, T and S are linearly dependent, which is absurd. This completes the proof of the theorem. $\square$

From now on, we shall always assume that Ω is a Reinhardt domain in $\mathbb{C}^2$ which is a unit ball with respect to some norm.

2 Candidate Partial Result

Theorem 1. *Let $1 \leq p, q < \infty$. If $\max\{p, q\} > 2$, then $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ does not embed linearly isometrically into S_∞ .*

This is a partial result toward Ray's conjecture. It extends the printed $2 < p, q < \infty$ consequence in Remark 4.2 to every pair with at least one exponent greater than 2.

3 Proof

We use the general criterion implicit in Ray's proof of Theorem 1.2 and made explicit in Remark 4.2. Let $X = (\mathbb{C}^2, \|\cdot\|)$. Suppose:

1. $\|(z_1, z_2)\| = \|(\bar{z}_1, \bar{z}_2)\|$;
2. for every $0 < c \leq 1$, the map $(z_1, z_2) \mapsto (z_1, cz_2)$ is not a contraction from X to ℓ_2^2 ;
3. the set $\{(z_1, z_2) : \|(z_1, z_2)\| > |z_1|\}$ is dense in $\mathbb{C}^2$;
4. $\|(1, 0)\| = \|(0, 1)\| = 1$;
5. $\|(1, z)\| \geq 1$ and $\|(w, 1)\| \geq 1$ for all $z, w \in \mathbb{C}$.

Then Ray's descent argument shows that X cannot embed isometrically into S_∞ . For completeness, we recall the mechanism. If $\Phi(z_1, z_2) = z_1 T + z_2 S$ were an isometry into compact operators, then, after the standard adjoint doubling and unitary conjugacy from Ray's proof, one may assume that T is positive diagonal and has a finite-dimensional one-eigenspace. Birkhoff–James orthogonality gives $\zeta \in M_T$ with $\langle T\zeta, S\zeta \rangle = 0$. Put $c = \|S\zeta\|_2$. Then

$$\|(z_1, z_2)\| \geq \|z_1 T\zeta + z_2 S\zeta\|_2 = (|z_1|^2 + c^2 |z_2|^2)^{1/2}.$$

If $c > 0$, this says that $(z_1, z_2) \mapsto (z_1, cz_2)$ is a contraction from X to ℓ_2^2 , contradicting item 2. If $c = 0$, the row and column deletion step in Ray's proof applies, using items 1 and 3, and produces a new isometric embedding with the multiplicity of the eigenvalue 1 of T reduced by one. After finitely many iterations T has norm strictly less than 1, contradicting item 4. Thus no such embedding exists.

It remains to check the criterion for $B_{p,q}$ when $q > 2$. Items 1, 4, and 5 are immediate from the Reinhardt unit ball. For item 3, if $z_2 \neq 0$ then $\|(z_1, z_2)\|_{B_{p,q}} > |z_1|$: otherwise, after scaling to the unit ball boundary, the first term already contributes at least 1 and the second contributes a positive amount. Hence the required set contains $\{z_2 \neq 0\}$, which is dense.

For item 2, fix $0 < c \leq 1$. Choose $y > 0$ and set

$$x = (1 - y^q)^{1/p}.$$

Then (x, y) lies on the $B_{p,q}$ -unit sphere. As $y \downarrow 0$,

$$x^2 + c^2 y^2 = (1 - y^q)^{2/p} + c^2 y^2 = 1 - \frac{2}{p} y^q + o(y^q) + c^2 y^2.$$

Since $q > 2$, the positive term $c^2 y^2$ dominates the order y^q loss for all sufficiently small y . Thus $x^2 + c^2 y^2 > 1$, so $(z_1, z_2) \mapsto (z_1, cz_2)$ is not contractive.

Therefore no isometric embedding exists when $q > 2$. If $p > 2$, apply the same argument after swapping the two coordinates, which identifies $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ isometrically with $(\mathbb{C}^2, \|\cdot\|_{B_{q,p}})$. This proves the theorem.

4 Verification Notes

The script `code/check_Bpq_obstruction.py` performs two sanity checks. First, for representative values with $q > 2$, it finds explicit small y witnesses to the non-contraction condition above. Second, it checks the known triangular positive case $B_{1,2}$ using the closed-form operator norm of $\begin{pmatrix} b & a \\ 0 & b \end{pmatrix}$. These numerical checks are not part of the proof; they only guard against a sign or exponent mistake.

5 Limitations and Novelty Status

This packet does not settle Ray’s full conjecture. In particular, it does not address the remaining region $1 \leq p, q \leq 2$ except for cases already handled in the source paper. The novelty check is local to this run: the registry, solution folders, and attempt notes contained no prior entry for arXiv:1911.00241 or the $B_{p,q}$ conjecture. No broad external literature search was performed for this partial result.

Human review should focus on whether Ray’s Remark 4.2 is being used exactly as its proof supports, and on the elementary $q > 2$ non-contraction calculation.